

* Max Consecutive 1's Replace -:

Example -:

A = 0 1 1 1 0 1 1 0 1 1 0
 ↓ ↓ ↓ ↓
 4 6 5 3

ans = 6

A = 1 1 1 1 1 1
 ans = 6

Corner case -: when all elements are already 1
 Ans = Array Size

Idea -:

For every 0 in arr[]

- 1) Iterate left and get count of consecutive 1's $\rightarrow l$
- 2) Iterate right and get count of consecutive 1's $\rightarrow r$
- 3) $cnt = l + r + 1$
- 4) compute max of cnt

* Max Consecutive 1's (Swap) -:

Example -:

0 1 1 1 0 1 1 0 0
 0 1 1 1 0 1 1 0 0

ans = 9 Count = l + r No extra ones

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if (l + r == count of ones)
    no extra ones -> l + r
else
    count = l + r + 1
```

* Increasing Triplets -:

$i < j < k$ & $A[i] < A[j] < A[k]$

Example -:

A = 2 6 9 4 10

$i < j < k$	$A[i] < A[j] < A[k]$
0 1 2	2 6 9
0 1 4	2 6 10
0 2 4	2 9 10
0 3 4	2 4 10
1 2 4	6 9 10

ans = 5 triplets

Idea -:

For every $A[j]$, calculate $\rightarrow N$ iterations

$\rightarrow$ No. of elements $< A[j]$

from $[0, j-1] \rightarrow l$

$\rightarrow$ No. of elements $> A[j]$

from $[j+1, N-1] \rightarrow r$

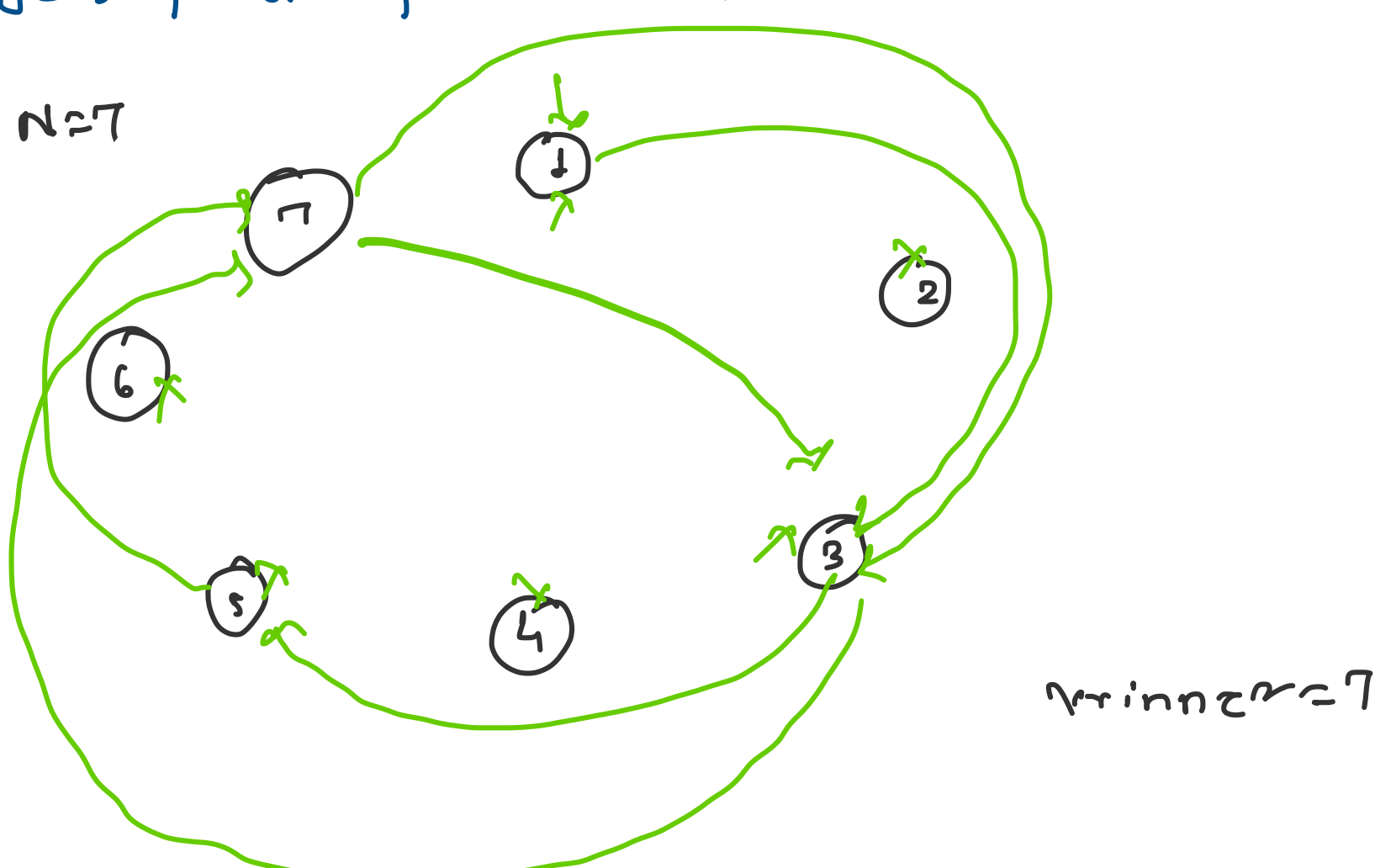
$\rightarrow$ Count $l + r$

$\rightarrow$ calculate sum of all count

Tc: $O(N^2)$

Sc: $O(1)$

* Josephus problem -:



Observation -:

If N is a power of 2, and we start from x , winner is x

$N=100$

100 people = $\frac{36 \text{ kills}}{36 \text{ pass}} \rightarrow 64 \text{ people}$

Ans = $2(36) + 1$

= 73

$N=16$

16 = $\frac{0 \text{ kills}}{0 \text{ pass}} \rightarrow 16$

ans = $2(0) + 1$

Tc: $O(\log_2 N)$

Sc: $O(1)$

$N \rightarrow$ Input

$x =$ closest power of 2

Kills = $N - x$

ans = $2(\text{Kills}) + 1$